1 Important Inequalities

The proofs of the error bound formulas all use the *Triangle Inequality*

$$\left|\sum_{i=1}^{n} x_i\right| \leq \sum_{i=1}^{n} |x_i|,$$

which you proved in AEP #1, as well as the following similar-looking inequality for integrals.

Proposition 1.1 (Triangle inequality for integrals)**.** *If $f : [a,b] \to \mathbb{R}$ is integrable then so is $|f|$ and*

$$\left|\int_a^b f(x)dx\right| \leq \int_a^b |f(x)|dx.$$

1. Assume the first statement in Proposition 1.1 and prove the second.
(*Hint*: $-|f(x)| \leq f(x) \leq |f(x)|$.)

2 The Trapezoidal Rule

Theorem 2.1 (Error bound for the trapezoidal approximation)**.** *Let $E_n^T(f)$ denote the absolute error $|\sum_{i=1}^n f(x_i^*)\Delta x - \int_a^b f(x)dx|$ using the trapezoidal approximation. If $M = \max\{|f''(x)|\}_{x\in[a,b]}$, then $E_n^T \leq \frac{K(b-a)^3}{12n^2}$.*

Proof. (a) Show that the error can be written as

$$E_n^T(f) = \left|\sum_{i=1}^n L_i\right|,$$

where

$$L_i := \left(\frac{f(x_{i-1})+f(x_i)}{2}\right)\frac{b-a}{n} - \int_{x_{i-1}}^{x_i} f(x)dx. \tag{2.2}$$

(b) Show that Equation (2.2) can be rewritten as

$$L_i = \int_{x_{i-1}}^{x_i} \left(t - \frac{x_{i-1} + x_i}{2} \right) f'(t) dt. \tag{2.3}$$

(*Hint*: Integrate Equation (2.3) by parts.)

(c) Integrate Equation (2.3) another way to show that

$$L_i = \frac{1}{2}\int_{x_{i-1}}^{x_i}\left(\left(\frac{b-a}{2n}\right)^2 - \left(t - \frac{x_{i-1}+x_i}{2}\right)^2\right) f''(t)dt.$$

(d) Now assume that $|f''(t)| \leq M$ for all $t \in [a,b]$ and finish the derivation.

□